

ME 333
MECHANICS OF MATERIALS

Lecture 10: Stress - Strain Relations

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Stress - Strain Relations
Introduction to stress-strain relations

- In general, for any material we can define stress as:

$$\sigma = f(\epsilon, \dot{\epsilon}, T, \dots)$$

Stress - Strain Relations

Stress - Strain relations

- According to the first law of Thermodynamics,

$$\delta W + \delta H = \delta U + \delta K$$

Note: Only changes in internal energy are measurable.

- Considering adiabatic (no heat flow) and static equilibrium conditions ($\delta H = 0$, $\delta K = 0$):

$$\delta W = \delta U$$

- Consider (u, v, w) be displacements for each point in a deformed member. Then:

$$\delta \epsilon_{xx} = \frac{\partial \delta u}{\partial x}; \quad \delta \epsilon_{yy} = \frac{\partial \delta v}{\partial y}; \quad \delta \epsilon_{zz} = \frac{\partial \delta w}{\partial z}$$

$$\delta \epsilon_{xy} = \frac{1}{2} \left(\frac{\partial \delta v}{\partial x} + \frac{\partial \delta u}{\partial y} \right); \quad \delta \epsilon_{yz} = \frac{1}{2} \left(\frac{\partial \delta w}{\partial z} + \frac{\partial \delta v}{\partial y} \right); \quad \delta \epsilon_{xz} = \frac{1}{2} \left(\frac{\partial \delta u}{\partial z} + \frac{\partial \delta w}{\partial x} \right)$$

Stress - Strain Relations

Stress - Strain relations

- Consider any arbitrary point P on a surface S , then:

$$\sigma_P = \sigma_{Px} \hat{i} + \sigma_{Py} \hat{j} + \sigma_{Pz} \hat{k}$$

- Work done due to surface tractions:

$$\begin{aligned} \delta W_S &= \int_S \sigma_{Px} \delta u \, dS + \int_S \sigma_{Py} \delta v \, dS + \int_S \sigma_{Pz} \delta w \, dS \\ &= \int_S (\sigma_{xx} l + \sigma_{xy} m + \sigma_{xz} n) \delta u \, dS + \int_S (\sigma_{yx} l + \sigma_{yy} m + \sigma_{yz} n) \delta v \, dS \\ &\quad + \int_S (\sigma_{zx} l + \sigma_{zy} m + \sigma_{zz} n) \delta w \, dS \end{aligned}$$

- Work done due to body forces:

$$\delta W_B = \int_V (B_x \delta u + B_y \delta v + B_z \delta w) \, dV$$

Stress - Strain relations

- Using divergence theorem:

$$\int_V (\nabla \cdot \mathbf{F}) dV = \oint_S (\mathbf{F} \cdot \mathbf{N}) dS$$

- Total work done is:

$$\begin{aligned} \delta W &= \delta W_S + \delta W_B \\ &= \int_V \left[\frac{\partial}{\partial x} (\sigma_{xx} \delta u + \sigma_{xy} \delta v + \sigma_{xz} \delta w) + \frac{\partial}{\partial y} (\sigma_{yx} \delta u + \sigma_{yy} \delta v + \sigma_{yz} \delta w) + \dots \right] dV \\ &\quad + (B_x \delta u + B_y \delta v + B_z \delta w) dV \\ &= \int_V [\sigma_{xx} \delta \epsilon_{xx} + \sigma_{yy} \delta \epsilon_{yy} + \sigma_{zz} \delta \epsilon_{zz} + 2\sigma_{xy} \delta \epsilon_{xy} + \dots] dV \end{aligned}$$

Stress - Strain relations

- The internal energy can be considered as:

$$U = \int_V U_0 dV \implies \delta U = \int_V \delta U_0 dV$$

- The internal strain energy depends on:

$$U_0 = U_0(\epsilon_{xx}, \dots, x, y, z, t)$$

- The variation in displacements leads to variation in internal energy as:

$$\delta U_0 = \frac{\partial U_0}{\partial \epsilon_{xx}} \delta \epsilon_{xx} + \frac{\partial U_0}{\partial \epsilon_{yy}} \delta \epsilon_{yy} + \dots + \frac{\partial U_0}{\partial \epsilon_{xy}} \delta \epsilon_{xy} + \dots$$

Stress - Strain relations

- From above relations:

$$\begin{aligned} \delta W &= \int_V [\sigma_{xx} \delta \epsilon_{xx} + \sigma_{yy} \delta \epsilon_{yy} + \sigma_{zz} \delta \epsilon_{zz} + 2\sigma_{xy} \delta \epsilon_{xy} + \dots] dV \\ \delta U &= \int_V \left[\frac{\partial U_0}{\partial \epsilon_{xx}} \delta \epsilon_{xx} + \frac{\partial U_0}{\partial \epsilon_{yy}} \delta \epsilon_{yy} + \dots + \frac{\partial U_0}{\partial \epsilon_{xy}} \delta \epsilon_{xy} + \dots \right] dV \end{aligned}$$

- Using work and energy equivalence:

$$\delta W = \delta U$$

we obtain:

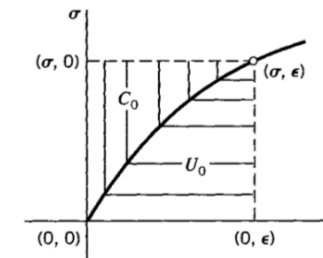
$$\sigma_{xx} = \frac{\partial U_0}{\partial \epsilon_{xx}}, \dots, \sigma_{xy} = \frac{1}{2} \frac{\partial U_0}{\partial \epsilon_{xy}}, \dots$$

Strain Energy & Complementary Energy

- The strain energy and complementary energy can be defined as:

$$U_0 = \int \sigma d\epsilon \implies \sigma = \frac{dU_0}{d\epsilon}; \quad C_0 = \int \epsilon d\sigma \implies \epsilon = \frac{dC_0}{d\sigma}$$

- Total area measure for a given stress strain pair is: $U_0 + C_0 = \sigma \epsilon$



Ref Fig: A. P. Boresi and R. J. Schmidt, Advanced Mechanics of Materials, Sixth Edition.

Reading Assignment

Chapter 3: Section 3.1 - 3.3.2